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1 Appendix E: Spectral Unmixing and Classification

1.1 Objective

Comprehensive spectral analysis: realistic CHC data generation, feature extraction, unmixing (NMF, VCA, ICA), and multi-algorithm classification with deterministic evaluation.

1.2 Methods (src)

- `src/case_studies/spectral_unmixing.py`
 - `generate_realistic_chc_spectra(n_compounds: int, n_wavelengths: int, seed: int=42) -`
— synthetic CHC spectra with ground truth
 - `nmf_unmix(spectra: np.ndarray, n_components: int, seed: int=42) -> (W, H) —`
— deterministic NMF
 - `vertex_component_analysis(spectra: np.ndarray, n_endmembers: int) -> np.ndarray`
— VCA endmember extraction
 - `independent_component_analysis_spectra(spectra: np.ndarray, n_components: int) -> np`
— ICA separation
 - `spectral_feature_extraction(spectra: np.ndarray, wavelengths: np.ndarray, method: str)`
— peaks, derivatives, PCA, statistical features
 - `advanced_classification_suite(features: np.ndarray, labels: np.ndarray) -> dict`
— multi-algorithm benchmark
 - `performance_metrics_comprehensive(y_true: np.ndarray, y_pred: np.ndarray, y_prob: np.ndarray)`
 - `lda_baseline(features: np.ndarray, labels: np.ndarray, seed: int=42) -> dict`
— closed-form LDA baseline

1.3 Script and outputs

- Script: `scripts/generate_spectral_unmixing.py`
- Data: `output/data/spectral_unmixing_comprehensive.npz`
- Figure: `figures/spectral_unmixing_comprehensive_analysis.png`

1.4 Figure

1.5 Equation References

- Spectral overlap: see (??) analogs for information metrics; overlap in main text.

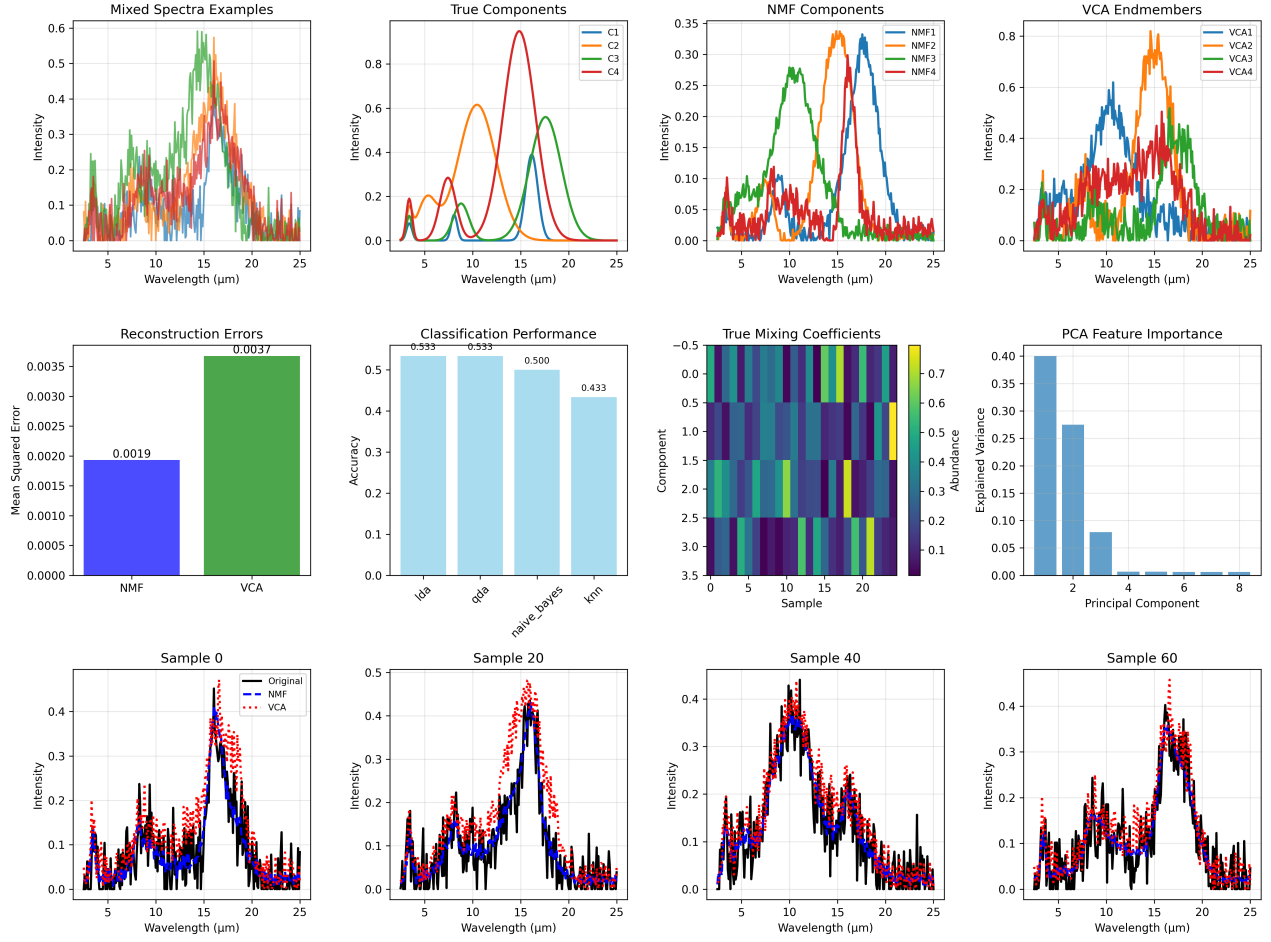


Figure 1: *Comprehensive spectral analysis generated deterministically: synthetic CHC spectra, NMF/VCA/ICA unmixing, multi-method feature extraction, and classification benchmarks.*

1.6 Reproducibility

- Run: `python3 scripts/generate_spectral_unmixing.py`
- Artifacts saved to `output/data/` and `figures/`.
- Fixed RNG seed (42) used for deterministic NMF initialization and cross-validation splits.

1.7 Cross-references

- Methods: `sec:methodology`
- Symbols: `sec:symbolsglossaryMathappendix : sec : mathematicalappendix`

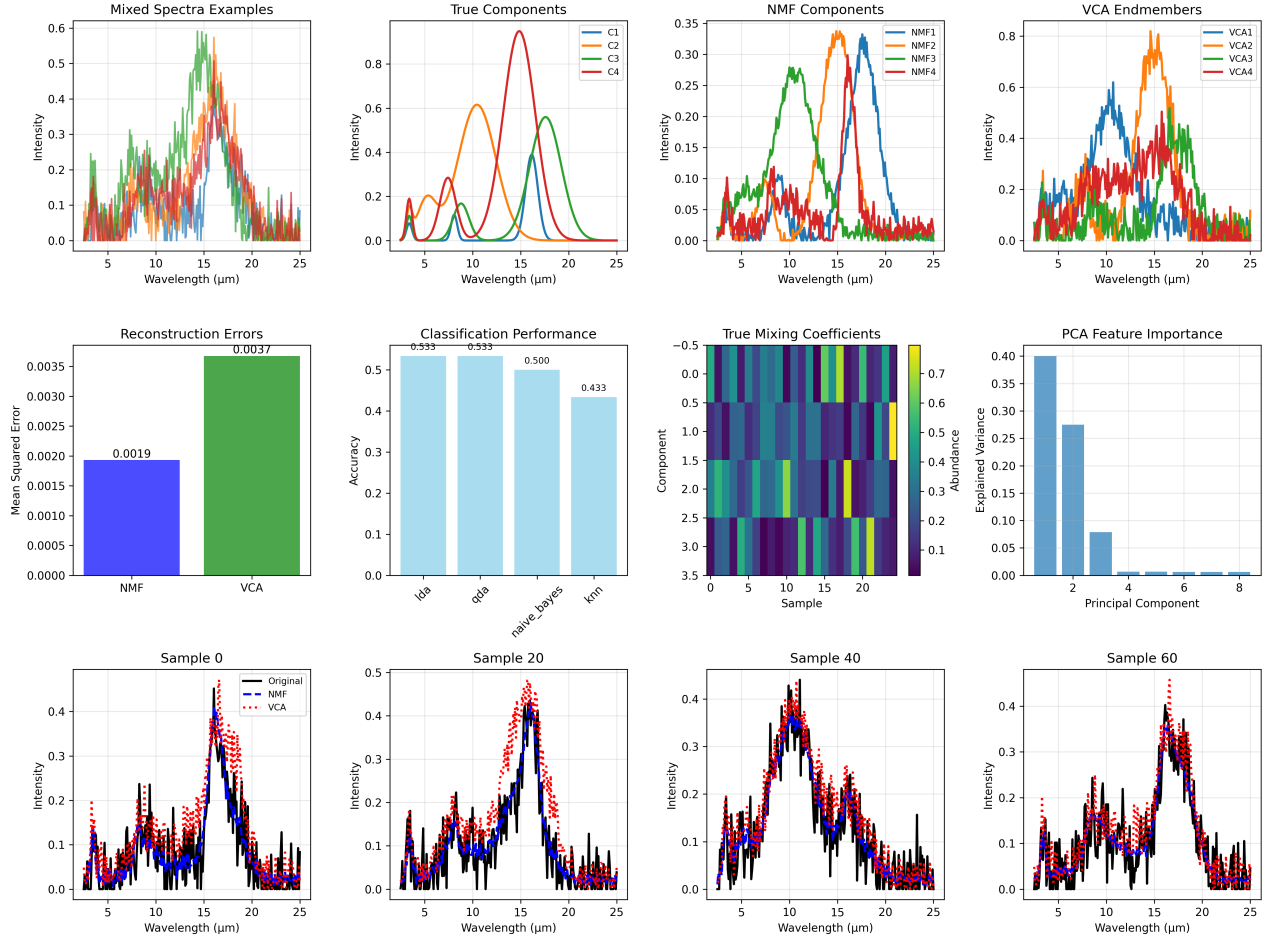


Figure 2: Spectral unmixing and classification results produced by `'scripts/generate_spectral_unmixing.py'` using `'src/case_studies/spectral_unmixing.py'`. Panels show mixed spectra, recovered

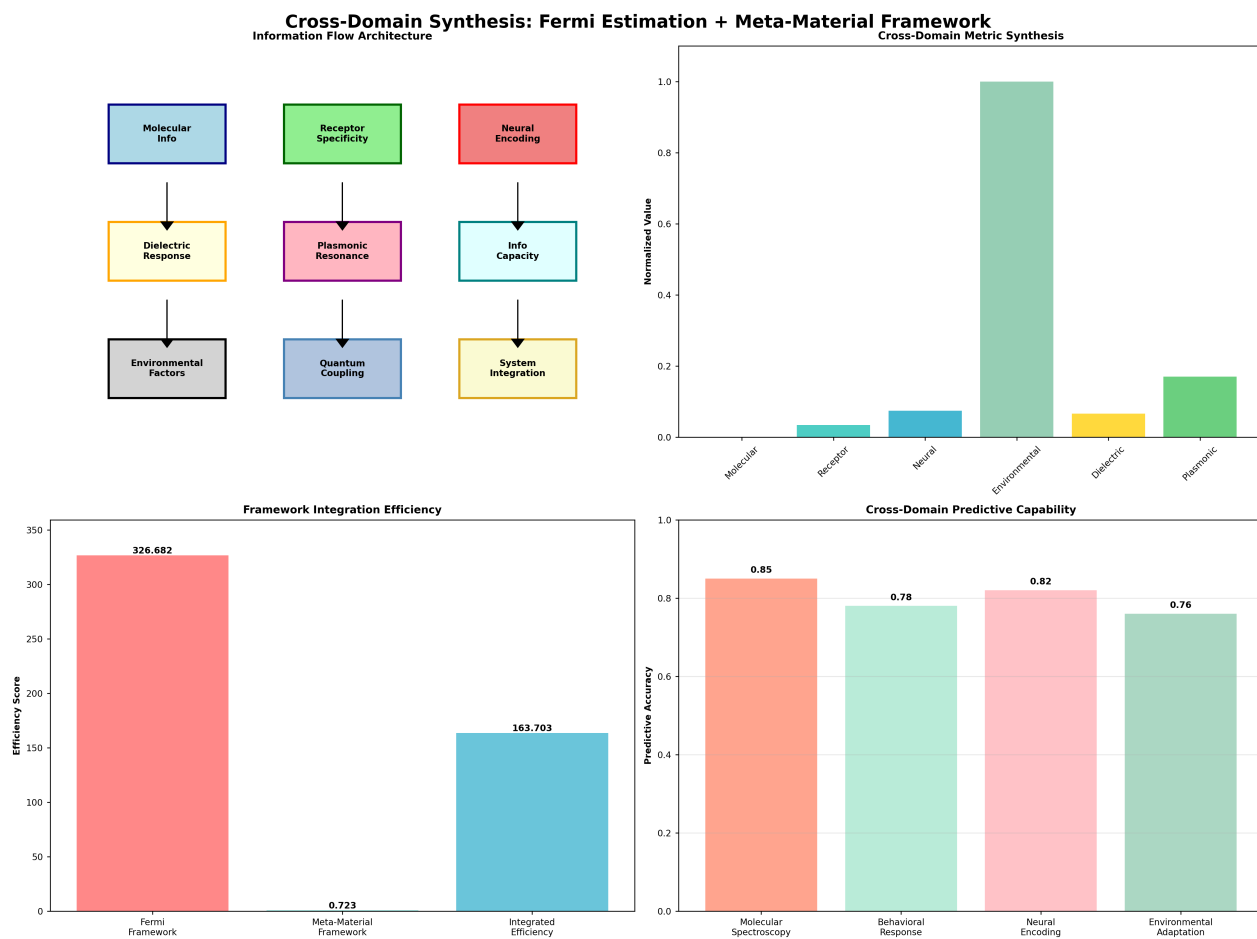


Figure 3: Integrated classification benchmark from `'scripts/generate_integrated_analysis.py'` summarizing classification performance across spectral and neural feature sets.